Module 2, Lesson 2.4: Understanding Memory & Context

Why Memory Changes Everything

Imagine working with a brilliant assistant who forgets everything you've ever told them the moment you leave the room. Every conversation starts from zero. Every preference needs to be re-explained. Every project context must be rebuilt.

That's what AI without memory looks like.

Now imagine an assistant who remembers your preferences, your projects, your past decisions, and the nuances of how you like to work. Each conversation builds on the last. Knowledge accumulates. Context deepens. The relationship matures.

That's what AI with memory enables—and it transforms everything.

How AI Memory Works

OpenClaw implements memory through a layered system that mimics how humans handle information.

Layer 1: Conversation Context (Short-Term Memory)

Within a single session, your agent maintains context about:

- What you've discussed
- What decisions have been made
- What's been accomplished
- What's still pending

This happens automatically. When you say "Do the same thing for my other clients," your agent knows what "the same thing" means because it remembers the context of

the current conversation.

Example:

```
You: "Create a project outline for my AI automation consulting."

Agent: [Creates outline]

You: "Add a section on pricing models."

Agent: [Adds section - knows which outline you mean]

You: "Make the third section more detailed."

Agent: [Expands section - knows which section is third]
```

This context persists until the session ends.

Layer 2: Daily Notes (Medium-Term Memory)

Each day, OpenClaw creates a file in the `memory/` directory:

- `memory/2024-01-15.md`
- `memory/2024-01-16.md`
- etc.

These files capture:

- What you worked on that day
- Decisions made
- Tasks started or completed
- Important information discussed

When you start a session, your agent loads recent daily notes to regain context from previous sessions.

What gets stored in daily notes:

- Summaries of conversations

- Tasks and their status
- Important facts you mentioned
- Insights or decisions

What doesn't get stored:

- Casual chitchat
- Repeated information already in memory
- Temporary debugging or exploration
- Confidential details you ask to exclude

Layer 3: MEMORY.md (Long-Term Memory)

The `MEMORY.md` file stores persistent information that should survive across all sessions:

- Key facts about you and your work
- Preferences you've established
- Important decisions and their reasoning
- Ongoing projects and their status
- Lessons learned

This is your agent's "permanent record" of what matters.

What belongs in MEMORY.md:

- Your goals and priorities
- Key contacts and their context
- Systems and processes you've established
- Preferences that should always be remembered
- Important dates and deadlines
- Insights worth preserving

How to add to MEMORY.md:

- Direct editing: Open the file and add information

- Through conversation: “Remember this: [information]”
- Automatic: Your agent can update it based on interactions

Layer 4: Configuration Files (Structural Memory)

Your configuration files (`USER.md`, `AGENTS.md`, `IDENTITY.md`, `SOUL.md`) provide structural memory about:

- Who you are (USER.md)
- How the agent should behave (AGENTS.md)
- The agent’s identity (IDENTITY.md)
- The agent’s personality (SOUL.md)

These don’t change frequently but provide foundational context for every interaction.

Working with Memory Effectively

Being Intentional About What Gets Remembered

Not everything should be in permanent memory. Being selective improves quality:

Do remember:

- Important decisions and their reasoning
- Preferences that should persist
- Key relationships and their context
- Systems and processes you’ve established
- Insights and lessons learned

Don’t remember:

- Temporary states (what you had for lunch)
- Information that will quickly become outdated
- Details better stored elsewhere (in files, calendars, etc.)
- Things you might want to change later

Memory Commands

You can explicitly manage memory through conversation:

"Remember this:" Adds information to long-term memory. "Remember this: My weekly team meeting is every Tuesday at 10 AM. I need to prepare updates the day before."

"Forget:" Removes information from memory. "Forget my old phone number and remember my new one: 555-123-4567."

"What do you know about:" Queries what's in memory. "What do you know about my consulting business?"

"Update memory:" Modifies existing information. "Update memory: The weekly meeting is now at 2 PM, still on Tuesdays."

"Don't remember:" Flags information as temporary. "Don't remember this conversation, but help me with…"

Reviewing and Maintaining Memory

Periodically review your agent's memory to ensure quality:

Weekly Memory Review: "What have you learned about me this week? Summarize what you've added to memory."

Monthly Memory Audit: "Let's review what you know about me. Read through MEMORY.md and identify anything outdated or incorrect."

Quarterly Memory Clean-up: "Help me clean up my memory file. Archive completed projects, remove outdated information, and highlight anything we should add."

Memory in Action: A Real Example

Let's see how memory builds over time:

Day 1:

```
You: "I'm starting an AI automation consulting business. My target market is small businesses with 5-50 employees who are struggling with operational inefficiencies."

Agent: "I've noted that. You're launching an AI automation consulting business targeting small businesses with 5-50 employees who have operational inefficiencies. What services are you planning to offer?"

[Agent adds this to daily notes and MEMORY.md]
```

Day 5:

```
You: "I've decided to focus on three service offerings: process automation, AI implementation, and training."

Agent: "Great, I'll add those to your business profile. You're offering process automation, AI implementation, and training for your target market of small businesses with 5-50 employees."

[Agent updates MEMORY.md with services]
```

Day 12:

```
You: "I closed my first client! Acme Corp, 25 employees, manufacturing company. We're starting with process automation for their inventory system."

Agent: "Congratulations! I'll record this milestone. Acme Corp is your first client—a 25-employee manufacturing company. You're starting with inventory system process automation. How long is the engagement?"

[Agent adds client to MEMORY.md under clients section]
```

Day 30:

```
You: "What clients do I have right now and what are we doing for each?"

Agent: "Based on my records:
1. Acme Corp (25 employees, manufacturing) - Inventory process automation,
ongoing
2. Smith & Associates (15 employees, law firm) - AI implementation for
document review, starting next week
3. Your process automation service is your most requested offering, followed
by training.

Would you like more details on any of these?"

[Agent recalls from MEMORY.md without needing to be re-informed]
```

This progression shows how memory creates continuity. By Day 30, you don't need to re-explain your business, services, or clients. The context accumulates.

Context Loading: How Your Agent "Wakes Up"

When you start a new session, your agent loads context in this order:

1. **Configuration files** - Your identity, the agent's identity, rules
2. **MEMORY.md** - Long-term memory
3. **Recent daily notes** - Usually the last 2-7 days
4. **Current session context** - Builds as you interact

This is why your agent might ask "What would you like to work on?" rather than immediately diving into a project—it doesn't know what happened since its last memory update.

Optimizing Context Loading

Help your agent load the right context:

At session start: "I want to continue working on [specific project]."

When switching contexts: "Let's shift gears to [different project]. Here's where we left off…"

When returning after time away: “It’s been a week. Let me get you up to speed on what’s happened with…”

Memory Limitations and How to Work Around Them

Limitation 1: Memory Size

Your agent can only load so much context at once. If MEMORY.md grows too large, earlier information may be less accessible.

Solution: Periodically summarize and condense. Archive old details, keep key insights.

Limitation 2: Memory Accuracy

Your agent remembers what you told it, but might misinterpret or store things imprecisely.

Solution: Verify important memories. “What do you have recorded about [important thing]?”

Limitation 3: Memory Recency

Recent interactions are more accessible than older ones.

Solution: Move important information to MEMORY.md. It’s always loaded. Daily notes rotate out.

Limitation 4: Context Confusion

When working on multiple projects, context can get mixed.

Solution: Be explicit about which project you’re discussing. “For the Acme Corp project…”

Advanced Memory Techniques

Technique 1: Project-Specific Memory Files

For major projects, create dedicated memory files:

```
~/.openclaw/workspace/projects/acme-corp/PROJECT.md
```

Reference these when working on specific projects: “I’m working on Acme Corp. Read the project memory file first.”

Technique 2: Memory Tags

Use tags to categorize memories:

```
[WORK] Weekly meeting moved to Wednesday
[PERSONAL] Doctor appointment Friday 3 PM
[PROJECT:Acme] Deliverable due next Tuesday
```

Then query by tag: “Show me all PROJECT:Acme memories”

Technique 3: Memory Summaries

Periodically create summary files:

```
~/.openclaw/memory/monthly-summaries/2024-01-summary.md
```

Archive the month’s daily notes while keeping key insights accessible.

Coming Up Next

In Lesson 2.5, we’ll explore customizing your agent’s personality—making it truly feel like YOUR assistant rather than a generic AI.

Key Takeaways

1. **Memory creates continuity** - Without it, every session starts from zero
2. **Layered memory mimics human cognition** - Short-term, medium-term, and long-term
3. **Be intentional about what gets remembered** - Quality over quantity
4. **Review and maintain memory regularly** - Keep it accurate and relevant
5. **Work around limitations with structure** - Project files, tags, and summaries help